

Deep Learning Methods for Tiger Video Generation

Your Name

1 Data

1.1 Tiger Image Dataset

To generate realistic tiger videos, high-resolution images of tigers are collected from wildlife footage and open-source datasets such as ImageNet-Vid and YouTube-VIS. Images include frontal, profile, and tilted views with diverse lighting conditions.

Preprocessing steps:

- Resize all images to 128×128 pixels,
- Normalize pixel values to fit model distribution,
- Data augmentation: flipping, color jitter, and rotation to increase variability.

1.2 Motion Data (Video Clips)

Videos of tigers exhibiting different movements (walking, running, turning) are extracted. Optical flow or keypoint-based motion representations are computed to provide temporal guidance.

1.3 Audio (Optional)

If considering sound-driven motion (e.g., roar lip-sync), audio is processed into Mel-Spectrograms and temporally aligned with video frames.

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2 Network Architecture

2.1 GAN Baseline: MoCoGAN-style

The baseline framework uses a GAN to generate video clips from a random noise vector.

Generator:

- Inputs: content latent vector z_c (appearance) and motion latent vector $z_m(t)$ (temporal dynamics),
- 3D convolutional layers or ConvLSTM to generate video sequences.

Discriminator:

- 2D CNN for per-frame realism,
- 3D CNN for temporal consistency.

2.2 Diffusion-based Video Generator (Main)

Diffusion models generate temporally coherent tiger videos via iterative denoising in latent space.

Architecture:

- Input: random noise video clip + motion condition,
- 3D U-Net with temporal attention for denoising,
- Optional latent VAE encoder/decoder for efficiency.

Losses:

- Score matching loss for denoising,
- Perceptual loss to preserve texture and details,
- Optional temporal consistency loss.

2.3 Joint Motion Conditioning

For more controllable outputs, motion embeddings extracted from optical flow or pose keypoints can be injected into intermediate layers via adaptive normalization, ensuring realistic movement while preserving identity.

3 Resources & Infrastructure

3.1 Pre-trained Models & Tools

- GAN: MoCoGAN pre-trained weights (optional),
- Diffusion: pre-trained Video Diffusion Model or VAE backbone,
- Motion extractor: RAFT or keypoint detector,
- Audio: Librosa for Mel-Spectrogram extraction.

3.2 Hardware & Data Pipeline

High-performance GPUs (e.g., NVIDIA A100 or RTX 4090) are recommended. Efficient data loaders (PyTorch DataLoader) support large-scale video batches.

3.3 Evaluation Metrics

- Video quality: FVD, PSNR, SSIM, LPIPS,
 - Identity preservation: cosine similarity via pre-trained recognition model (ArcFace),
 - Temporal consistency: Average Pose Distance (APD) or frame-to-frame motion correlation.
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4 Design Philosophy & Thoughts

- **Modular design:** Separate GAN baseline and diffusion modules for clear comparison.
- **Temporal consistency:** Diffusion with temporal attention ensures smooth motion.
- **Motion-content disentanglement:** Motion conditions are injected at intermediate layers, preserving spatial fidelity.
- **Stage-wise training:** Pre-train modules individually before joint fine-tuning.
- **Future extensions:** Text- or audio-conditioned tiger generation for controllable video synthesis.